MCQ solutions

1. **B.** $1 \times 2 + 2 \times 0 - 1 \times 3 = -1$. The scalar product is a real number, not a vector.
2. **D.** The sum of the squares is 9; the norm is its square root, that is 3.
3. **A.** Two non-collinear directions generate the direction of the plane. A single test leaves directions not perpendicular to the vector.
4. **C.** The cosine of the angle is negative since the norms are positive. The non-oriented angle is therefore between $\pi/2$ and π.
5. **B.** Orthogonality concerns their directions. Perpendicular lines, moreover, intersect.
6. **D.** The point belongs to the plane and the vector that connects it to M is vertical, so normal to the plane.
7. **A.** The triangle MHP is right-angled at H when $P \neq H$; the identity remains valid for $P = H$. It proves $MP \geq MH$.
8. **C.** The volume is one third of the base area times the height: $6 \times 4/3 = 8$.
9. **B.** The scalar product is the cosine of the angle since the norms are 1; the non-oriented angle is $2\pi/3$.
10. **D.** Write $H = A + t(1;1;0)$. Orthogonality $(M - H) \cdot (1;1;0) = 1 - 2t = 0$ gives $t = 1/2$.

True/false. False: the projection gives the minimum, by Pythagoras. False: $(1;0;0)$ and $(0;1;0)$. True: each height must correspond to the chosen base.

Solutions to the applications and guidance

Application 1. Product 1, norms $\sqrt{2}$ and $\sqrt{2}$, therefore cosine $1/2$, angle $\pi/3$. The coordinate formula requires an orthonormal coordinate system.

Application 2. In the plane $z = 0$, $\vec{\imath} \cdot \vec{\jmath} = 0$, but $\vec{\jmath}$ lies in the plane and is not normal. Lines orthogonal lines have orthogonal directions; they are perpendicular if they also intersect.

Application 4. $H = (t; t; 0)$ and $\overrightarrow{MH} \cdot (1; 1; 0) = 2t - 5 = 0$, therefore $H = (5/2; 5/2; 0)$. $MH^2 = 1/4 + 1/4 + 1 = 3/2$; distance $\sqrt{6}/2$.

Application 5. $H = (1; 2; 0)$; $\overrightarrow{HM} = (0; 0; 3)$ is orthogonal to both directions $\vec{\imath}, \vec{\jmath}$. For $Q = (x; y; 0)$, $MQ^2 = (x-1)^2 + (y-2)^2 + 9 \geq 9$, equality only at H.

Application 6. $AB = \sqrt{52}$, $AC = \sqrt{45}$, product 36. The cosine is 36 $/\sqrt{2340}$; the sine is $\sqrt{2340 - 1296}/\sqrt{2340} = 6\sqrt{29}/\sqrt{2340}$. The area is therefore 3 $\sqrt{29}$ and the altitude 12/ $\sqrt{29}$.

Application 7. Missing: MH orthogonal to the plane (or $\overrightarrow{MH}$ collinear with a normal). For $M = (1; 2; 3)$, H=(1;2;0) works but Q=(0;0;0), although in the plane, is not the projection.

Teaching notes

Have students find the altitude in the figure before calculating a volume. For calculating an angle, require checking cos $\theta \in [-1; 1]$. Keep an exact expression before approximation.

Application 3. The midpoint is I=(1;1;0) and a normal vector is (2;2;0). The locus is the plane through I with this normal. $MA^2 - MB^2 = 4x + 4y - 8$; hence x+y=2. For example (1;1;0) and (2;0;3) work. They are all points of the plane, with no restriction on z.

Teaching notes and solutions

Guidance for the teacher

Start with the three expressions of the scalar product. Orthogonal lines are not necessarily intersecting: "perpendicular" requires intersection. To prove that a line is orthogonal to a plane, choose two non-collinear directions. The general formula for the point-to-plane distance need not be memorised: prefer the projection. Cartesian equations used in the progression in S54 can be introduced via the vector condition, then formalised in S67.

Solutions to the activity

1. A point H of d is written $H = A + t\vec{u}$, therefore $H(1+t, t, 0)$ for $t \in \mathbb{R}$.

2. $\overrightarrow{MH} = (1+t, t-2, -3)$. The scalar product with $\vec{u} = (1,1,0)$ is $1 + t + t - 2 = 2t - 1$. Set to zero gives $t = \frac{1}{2}$ & $H(\frac{3}{2}, \frac{1}{2}, 0)$.

3. The coordinate system is orthonormal: $MH^2 = (\frac{3}{2})^2 + (-\frac{3}{2})^2 + (-3)^2 = \frac{27}{2}$. Hence $MH = \frac{3\sqrt{6}}{2}$.

4. For $P \in d$, $\overrightarrow{HP}$ is collinear with $\vec{u}$ and thus orthogonal to $\overrightarrow{MH}$. Pythagoras' theorem gives $MP^2 = MH^2 + HP^2 \geq MH^2$, with equality only if $P = H$. The calculated point is indeed the unique point on the line closest to M.

E1. In the orthonormal coordinate system, the scalar product is $2 \times 1 + (-1) \times 4 + 3 \times 0 = -2$.

E2. The scalar product is $3 \quad a + 2(-1) + 1 \times 4 = 3a + 2$. Orthogonality is equivalent to it being zero: $3a + 2 = 0$, therefore $a = -\frac{2}{3}$.

E3. The two direction vectors are non-collinear and both scalar products are zero. The midpoint is $I(1,1,1)$. The required locus is the plane through I and normal to $\overrightarrow{AB} = (2,2,2)$, hence with normal vector $(1,1,1)$.

E4. The two vectors are non-zero, with norms $\sqrt{2}$, and their scalar product is 1. So their angle $\theta \in [0, \pi]$ satisfies $\cos \theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$, i.e. $\theta = \frac{\pi}{3}$.

E5. The area of the base rectangle is $\frac{3 \times 4}{2} = 6$ cm^2. The altitude perpendicular to this base is 5 cm, so $V = \frac{6 \times 5}{3} = 10$ cm^3.

Solutions to homework DM11

1. $\overrightarrow{AB} = (-2,2,0)$, $\overrightarrow{AC} = (-2,0,2)$ and $\overrightarrow{BC} = (0,-2,2)$; the three norms equal $2\sqrt{2}$. The triangle is equilateral. Its height is $\sqrt{(2\sqrt{2})^2 - (\sqrt{2})^2} = \sqrt{6}$; its area is therefore $\frac{(2\sqrt{2})(\sqrt{6})}{2} = 2\sqrt{3}$.

2. The vectors $\overrightarrow{AB}, \overrightarrow{AC}$ are not collinear. Their scalar products with $(1,1,1)$ are respectively $-2 + 2 = 0$ and $-2 + 2 = 0$. The non-zero vector $\vec{n}$ is therefore normal to (ABC).

3. The projection of O lies on the line with direction vector $\vec{n}$ through O, therefore $H = (t,t,t)$. Its condition for belonging to the plane is $(t-2) + t + t = 0$, i.e. $t = \frac{2}{3}$. We get $H(\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$, which satisfies the two conditions for a projection.

4. $OH = \sqrt{3(\frac{2}{3})^2} = \frac{2}{\sqrt{3}}$. Taking the base ABC, the volume is $V = \frac{(2\sqrt{3})(\frac{2}{\sqrt{3}})}{3} = \frac{4}{3}$ units of volume. Another base: the triangle OAB is right-angled at O, area $\frac{2 \times 2}{2} = 2$; the line (OC) is perpendicular to the plane (OAB), and $OC = 2$. Thus $V = \frac{2 \times 2}{3} = \frac{4}{3}$.

Solutions to the textbook or past-paper work

Exercise 23. In the given orthonormal coordinate system, the scalar product is the sum of the products of the corresponding coordinates:

$$(-3{,}5)(-4) + 2 \times 5{,}5 + (-4)(-3) = 14 + 11 + 12 = 37,$$

$$\frac{2}{3}\left(-\frac{3}{4}\right) + \frac{15}{7}\frac{14}{3} + \left(-\frac{12}{11}\right)\left(-\frac{22}{4}\right) = -\frac{1}{2} + 10 + 6 = \frac{31}{2}.$$

Exercise 27. In each case, the two vectors $\vec{u}$ and $\vec{v}$ given span the plane and are not collinear. A non-zero vector $\vec{w}$ is normal if, and only if, it is orthogonal to each of them.

1. $\vec{w} \cdot \vec{u} = -132 + 312 - 180 = 0$ and $\vec{w} \cdot \vec{v} = -66 + 546 - 480 = 0$. The vector $\vec{w}$ is normal to the plane.
2. $\vec{w} \cdot \vec{u} = 20 + 6 - 26 = 0$, but $\vec{w} \cdot \vec{v} = 18 - 6 - 10 = 2 \neq 0$. It is not normal to the plane.
3. $\vec{w} \cdot \vec{u} = \frac{1}{2} + 6 - 7 = -\frac{1}{2} \neq 0$. This single nonzero product is enough to rule out normality.